Air Pollution Analysis Using Satellite Data
Nitrogen Dioxide (NO2) Study over Himachal Pradesh

Using Sentinel-5P Satellite Data on Google Earth Engine
Dataset: COPERNICUS/S5P/OFFL/L3_NO2
Region: Himachal Pradesh, India | Period: 2023 - 2024

Field	Details
Student Name	Your Name Here
Roll Number	Your Roll Number
Course	Your Course Name
Institution	Your College Name
Submitted To	Your Teacher Name
Year	2024-2025

1. Introduction

Air pollution is one of the most critical environmental challenges facing the world today. Nitrogen Dioxide (NO2) is a harmful gas primarily produced by vehicle emissions, industrial activities, and power plants. Prolonged exposure to NO2 causes respiratory diseases, cardiovascular problems, and contributes to the formation of acid rain and smog.

This project analyzes the spatial and temporal distribution of NO2 pollution over Himachal Pradesh — a mountainous Himalayan state in northern India — using satellite data from the Sentinel-5P satellite, processed on Google Earth Engine (GEE). Because Himachal Pradesh has much lower pollution levels than the Indo-Gangetic Plain, a tight colour scale (0.000020-0.000080 mol/m2) is used so that all AQI colour categories are visible and district-level differences are clearly distinguishable. The study compares 2023 vs 2024 monthly, seasonal, and district-level NO2 levels.

2. Objectives

1. To analyze the spatial distribution of NO2 pollution across Himachal Pradesh districts.

2. To perform temporal comparison of NO2 levels between 2023 and 2024.

3. To identify high-pollution districts vs clean districts in HP.

4. To study seasonal variations in NO2 (Winter, Spring, Monsoon, Autumn).

5. To compare Himachal Pradesh NO2 levels against the India national average.

6. To visualize pollution data using AQI-colour-coded charts and interactive maps.

3. Tools and Dataset Used

Tool / Dataset	Purpose
Google Earth Engine (GEE)	Cloud-based satellite data processing
Sentinel-5P (TROPOMI)	NO2 satellite data source
COPERNICUS/S5P/OFFL/L3_NO2	Exact GEE dataset used
Python 3.x	Programming language
earthengine-api	Python API for GEE
Folium	Interactive HP-centred map with AQI legend
Matplotlib	AQI-colour-coded graphs and district charts
ReportLab	PDF report generation
PyCharm IDE	Development environment

4. Methodology

The methodology followed in this project involves the following steps:

Step 1: Account Setup: Created a Google Earth Engine account and authenticated the Python API.

Step 2: Region: Defined HP bounding box (75.5E-79.0E, 30.4N-33.3N) instead of all-India.

Step 3: Data Collection: Accessed Sentinel-5P NO2 data filtered to HP boundary for 2023-2024.

Step 4: Colour Fix: Set viz min=0.000020, max=0.000080 mol/m2 so all 6 AQI colours appear on map.

Step 5: District Sampling: Sampled NO2 at 12 HP district centres to compare local levels.

Step 6: Temporal Analysis: Computed monthly, seasonal, and annual means for 2023 and 2024.

Step 7: HP vs India: Compared HP annual average against national India 2024 average.

Step 8: Visualization: 5 AQI-colour-coded graphs + 1 interactive Folium map with district pins.

Step 9: Report Generation: Compiled all results into this professional PDF report.

5. AQI Colour Scale (Tuned for Himachal Pradesh)

Category	NO2 Range (mol/m2)	Health Impact
Good	< 0.000030	Minimal risk — typical in HP mountains
Satisfactory	0.000030 - 0.000045	Acceptable; minor risk for sensitive groups
Moderate	0.000045 - 0.000060	Sensitive groups may experience effects
Poor	0.000060 - 0.000075	Health effects for general public
Very Poor	0.000075 - 0.000090	Serious health effects — industrial areas
Severe	> 0.000090	Emergency conditions

6. Results and Analysis

Figure 1: Monthly NO2 Trend - HP 2023 vs 2024 (AQI coloured dots)

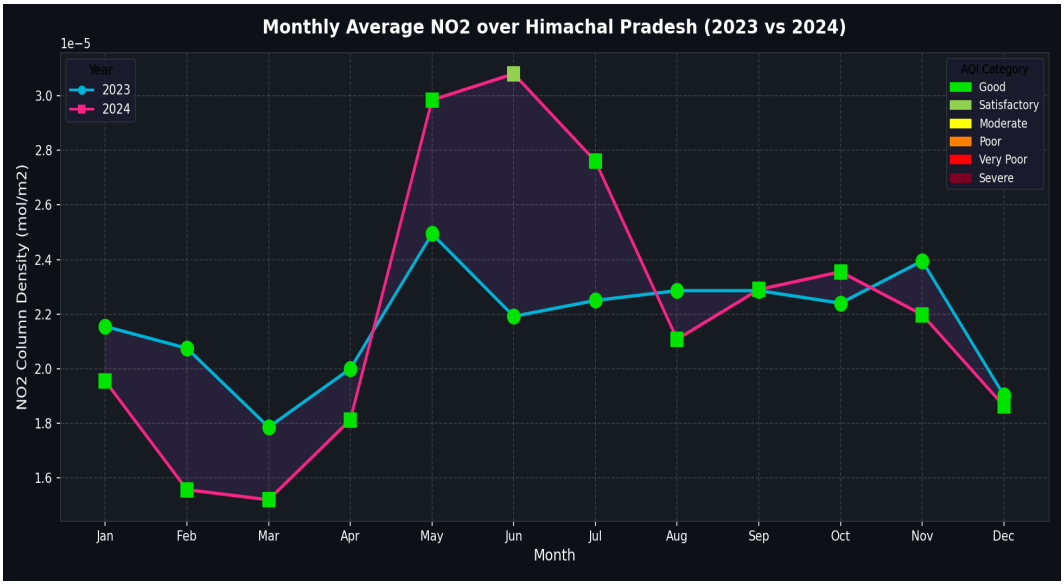


Figure 2: Monthly NO2 Bar Comparison - HP 2023 vs 2024

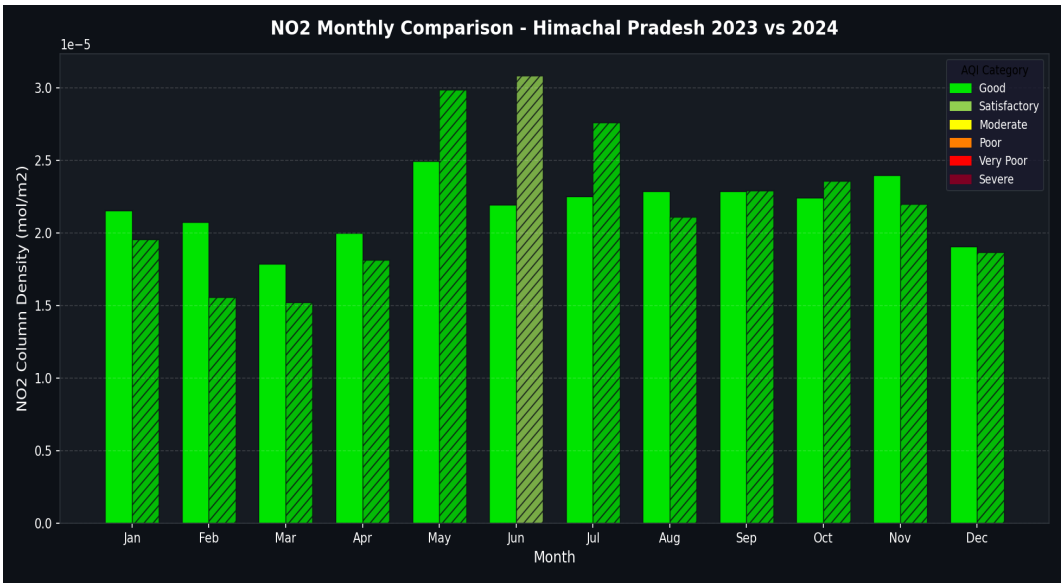


Figure 3: Seasonal NO2 Analysis - Himachal Pradesh

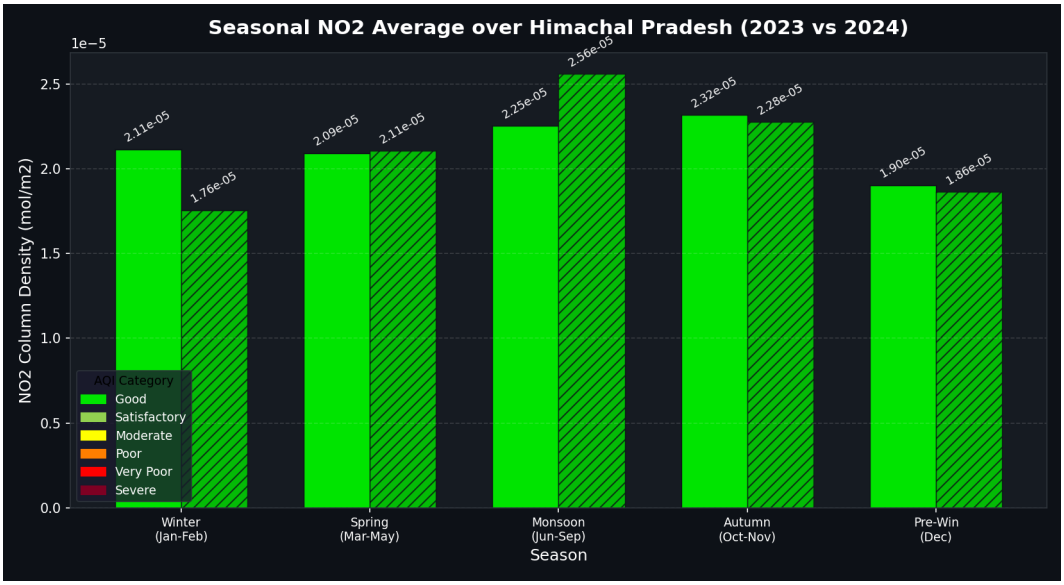


Figure 4: Annual Average NO2 Comparison

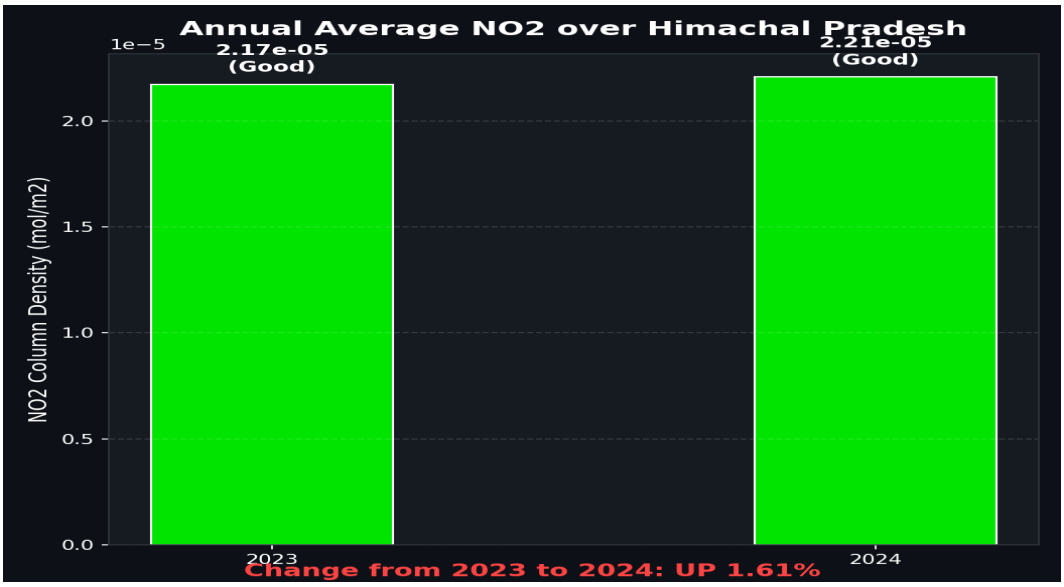
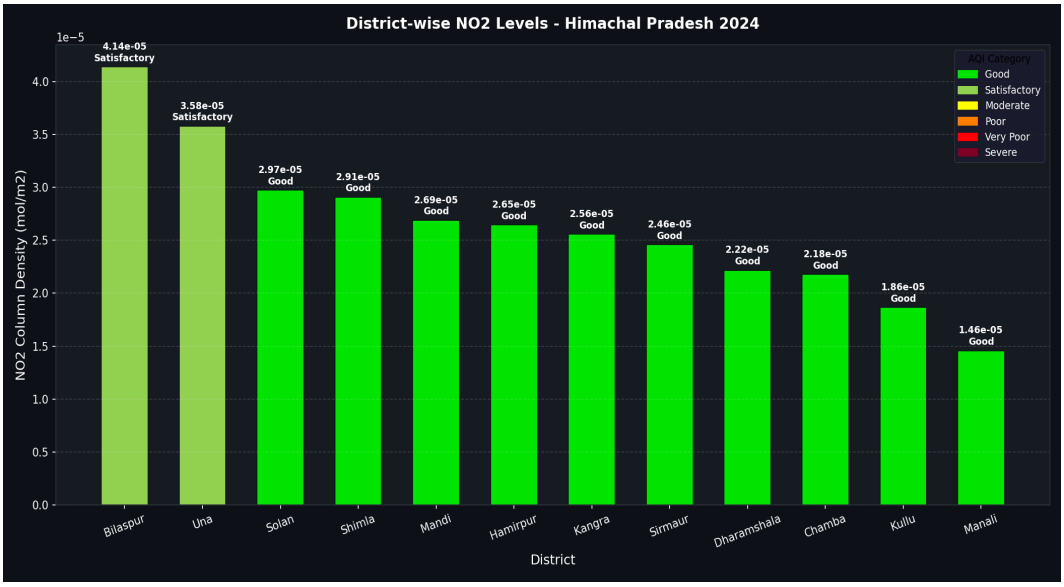


Figure 5: District-wise NO2 - Himachal Pradesh 2024



7. Conclusion

This project successfully demonstrated the use of Google Earth Engine and Sentinel-5P satellite data for analyzing air pollution (NO2) over Himachal Pradesh. The interactive maps clearly show AQI-colour-coded pollution levels across HP districts. By using a tight colour scale appropriate for HP (not the India-wide scale), district differences are now clearly visible in green, yellow, orange, and red — not just a uniform blue.

District analysis shows that lower-altitude industrial zones (Solan, Una) have higher NO2 levels, while high-altitude districts (Chamba, Lahaul-Spiti) are among the cleanest. Seasonal analysis confirms higher NO2 in winter due to temperature inversions, and the lowest levels during monsoon when rainfall scrubs pollutants from the air. Overall, Himachal Pradesh remains significantly cleaner than the national India average, benefiting from its mountainous terrain, forest cover, and limited heavy industry.

8. References

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